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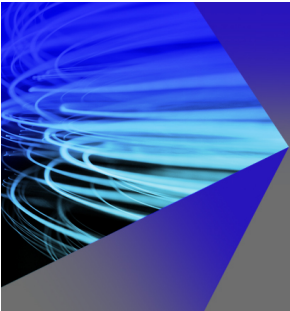
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ABSTRACT

Photodetachment cross sections (PDCSs) are key parameters for quantitatively characterizing the generation and annihilation of anions, with transition moment integrals being the critical component in their theoretical calculation. This work introduces an efficient method for calculating transition moments within the plane wave approximation, based on the analytical expressions of the recursive Cartesian two-center Gaussian integral. This method effectively bypasses the convergence issues commonly associated with traditional plane wave expansion techniques. By incorporating the random orientation of gas-phase molecular anions, the total PDCSs can be derived. For molecules with low anisotropy wavefunctions, the integration is further simplified to three orthogonal directions, dramatically reducing computational costs. Benchmark tests on a variety of atomic and molecular anions (H^- , Li^- , C^- , O^- , F^- , S^- , Cl^- , OH^- , NH_2^- , CN^- , and C_2H^-) show excellent agreement with previous experimental measurements and high-level theoretical results. Furthermore, the method's superior computational efficiency is demonstrated through applications to larger systems, such as benzyl anion, glycine anion, and water tetramer anion. The proposed method holds significant potential for advancing the field of vacuum ultraviolet photochemistry.

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I. INTRODUCTION

The electron attachment cross section and photodetachment cross section (PDCS) are crucial parameters for quantifying the photodestruction rates,^{1–3} relating to the generation and annihilation processes of anions in astrophysical and atmospheric environments.^{4–10} Experimental data for electron attachment cross sections are scarce, mainly because anions readily stabilize through collisions with inert gases. However, PDCS can provide insights into electron attachment cross sections, as the photodetachment process is reversible.^{11,12} This reversibility makes PDCS an important topic of recent theoretical interest on anions.¹³ In particular, the PDCS quantifies the probability that an anion absorbs a photon of a specific energy and ejects an electron. Calculating PDCS requires a detailed treatment of both the initial and final states involved in the

photodetachment process, with a particular emphasis on accurately modeling the continuum photoelectron wavefunction.¹⁴

By considering the interaction between the ejected photoelectron and the system, the explicit solution of the Schrödinger equation for the continuum wavefunction provides an accurate description of the photoelectron's behavior, especially for systems with resonances. However, obtaining the explicit solution requires complex algorithms and numerical techniques, such as Schwinger variational,^{15,16} complex Kohn,¹⁷ and R-matrix,^{18–22} which are computationally expensive, especially as the system size increases.

An alternative approach is to approximate the photoelectron wavefunction in the photodetachment process as a plane wave, assuming that the ejected photoelectron moves freely and neglecting interactions with the remaining core. Thus, this approximation is particularly accurate for photodetachment of anions because of

the remaining neutral molecule. Although the plane wave ansatz formally omits resonance effects and higher-order electron–core coupling beyond the Coulomb term, these omissions engender deviations that are well below the statistical uncertainty of the present calculations. By expanding the plane wave into spherical harmonics and radial functions, Krylov's research group successfully computed PDCS for atomic and small molecule anions using equation-of-motion coupled-cluster Dyson orbitals and reproduced the energy dependence.^{23,24} For convergent PDCS calculations, larger basis sets are often required to account for the dispersion characteristics of anion wavefunctions. Liu and Ning applied the density functional theory (DFT) level wavefunction calculations and obtained analytical calculations for random molecular orientations, reducing the computational cost for PDCS of large-system anions.²⁵

The plane wave expansion involves an infinite series for all angular momentum quantum numbers, leading to more complex integrals and Legendre polynomial calculations as the quantum number increases, which significantly raises the computational workload. Consequently, practical calculations usually truncate at a specific angular momentum quantum number. However, for systems involving high energy, large spatial scales, or high precision requirements, convergence issues with plane wave expansions may become problematic.²⁶

In this work, we derive an analytical expression for the transition dipole moment integrals between plane waves and Gaussian basis functions using Cartesian Gaussian integral recursive expressions. By incorporating this algorithm into the calculation of differential cross sections, the convergence issues associated with plane wave expansion techniques are effectively avoided. Benchmark tests on a series of atomic and molecular anions (H^- , Li^- , C^- , O^- , F^- , S^- , Cl^- , as well as molecular anions such as OH^- , NH_2^- , CN^- , and C_2H^-) were performed, with results compared with previous calculations and experimental measurements. In addition, the high efficiency of our method is also verified on larger anionic molecules (benzyl anion, glycine anion, and water tetramer anion). This study provides new insights into accurately modeling photodetachment processes in complex molecular anions.

II. THEORETICAL APPROACH

A. Doubly differential cross section

In the present work, we focus exclusively on one-photon photodetachment, which is closely associated with vacuum ultraviolet (VUV) light. Given that the wavelength of VUV light is much larger than the size of an atom or a molecule, the charge distribution during the photodetachment process can be approximated as an electric dipole. Under the linearly polarized light, the doubly differential cross section in the body-fixed frame is directly proportional to the square of the transition moment,^{16,27}

$$\frac{\partial^2 \sigma}{\partial \Omega_{\vec{k}} \partial \Omega_{\vec{\varepsilon}}} = \frac{4\pi^2 E}{c} |I_{\vec{k}, \vec{\varepsilon}}^L|^2 \quad (1)$$

for the dipole form (or the length gauge) and

$$\frac{\partial^2 \sigma}{\partial \Omega_{\vec{k}} \partial \Omega_{\vec{\varepsilon}}} = \frac{4\pi^2}{cE} |I_{\vec{k}, \vec{\varepsilon}}^V|^2 \quad (2)$$

for the velocity form (or the momentum gauge). E is the photon energy, $\vec{\varepsilon}$ represents the direction of polarization of the light, and $\vec{k}$ is the momentum of the photoelectron. $I_{\vec{k}, \vec{\varepsilon}}^{L,V}$ represents the transition moment for the dipole or velocity form. The transition moment connects the initial state wavefunction ψ_i and the final state wavefunction ψ_f during the photoionization process,

$$I_{\vec{k}, \vec{\varepsilon}}^L = \langle \psi_f | \vec{\varepsilon} \cdot \vec{r} | \psi_i \rangle = \langle \psi_{ele} \psi_{N-1} | \vec{\varepsilon} \cdot \vec{r} | \psi_N \rangle = \langle \psi_{ele} | \vec{\varepsilon} \cdot \vec{r} | \psi_{dyson} \rangle, \quad (3a)$$

$$I_{\vec{k}, \vec{\varepsilon}}^V = \langle \psi_f | \vec{\varepsilon} \cdot \vec{p} | \psi_i \rangle = \langle \psi_{ele} \psi_{N-1} | \vec{\varepsilon} \cdot \vec{p} | \psi_N \rangle = \langle \psi_{ele} | \vec{\varepsilon} \cdot \vec{p} | \psi_{dyson} \rangle. \quad (3b)$$

Here, $\vec{r}$ and $\vec{p} = -i\nabla$, respectively, present the length and the momentum operator. ψ_{ele} is the ejected photoelectron wavefunction, for the photodetachment process of anions, neglecting the interaction between the photoelectron and the neutral molecule, and ψ_{ele} is approximated as a plane wave $\psi_{\vec{k}}$.²⁸ ψ_{dyson} is the Dyson orbital connecting the initial N -electron state ψ_N and the final $N-1$ -electron state ψ_{N-1} ,

$$\psi_{dyson} = \sqrt{N} \int \Psi_{N-1}(x_2, \dots, x_N) \Psi_N(x_1, x_2, \dots, x_N) dx_2 \dots x_N. \quad (4)$$

The Dyson orbitals consider electron correlation and orbital relaxation effects during the ionization process.^{29–31} To satisfy the orthogonality of molecular orbitals, the origin of the Dyson orbitals is set to coincide with the laboratory coordinate origin, making the Dyson orbitals orthogonal to the plane wave.

Notably, Brumboiu *et al.* introduced a novel approach for calculating atomic cross sections that extends beyond the dipole approximation and systematically assesses the errors arising from the dipole approximation.³² Their results indicate that the corrections are generally insignificant in the VUV band, which is where the majority of the test systems in this study are evaluated. Consequently, all results presented here are calculated using the dipole approximation.

B. Recursive calculation of two-center Gaussian integrals

In first-principles computations utilizing the linear combination of atomic orbitals (LCAO) method, Gaussian functions are commonly used as basis functions to construct atomic orbitals. According to the scheme proposed by Obara and Saika,³³ Gaussian integrals can be solved recursively,

$$\langle a + 1_i | c | b \rangle = \frac{1}{2\alpha_\mu} \frac{\partial}{\partial A_i} \langle a | c | b \rangle + \frac{1}{2\alpha_\mu} N_i(a) \langle a - 1_i | c | b \rangle. \quad (5)$$

Here, a , b , and c represent the nominal angular momentum numbers of the Gaussian functions, where $N_i(n)$ is meant to take the value of the i th component of the nominal angular momentum index n . A_i is the i th component of the nuclear coordinate, $A_i = \{x_a, y_a, z_a\}$. α_μ is the exponent of the left-sided Gaussian function.

Based on Eq. (5), the following recursive relationships can be derived:

$$(a+1|x|b) = (G_x - x_A)\langle a|x|b \rangle + \frac{a}{2\alpha_T}\langle a-1|x|b \rangle + \frac{b}{2\alpha_T}\langle a|x|b-1 \rangle + \frac{1}{2\alpha_T}\langle a|b \rangle, \quad (6a)$$

$$(a+1|b) = (G_x - x_A)\langle a|b \rangle + \frac{a}{2\alpha_T}\langle a-1|b \rangle + \frac{b}{2\alpha_T}\langle a|b-1 \rangle, \quad (6b)$$

where G_x and α_T are the center and exponent of the new Gaussian function, respectively, and their explicit expression is provided in the [supplementary material](#) [see Eqs. (S-7) and (S-8)].

Similar to normal Gaussian functions, the plane wave function $e^{i\mathbf{k}\cdot\mathbf{r}}$ can also couple with Gaussian functions, resulting in a new Gaussian function although its center and exponent lie in the complex space. The detailed expressions are provided in the [supplementary material](#). This approach provides an efficient solution for transition moment integrals, applicable to both the length and momentum gauges.

C. Molecular orientation average

In experiments measuring the photodetachment cross sections of anions, the orientation of gas phase molecules is random. Therefore, in theoretical calculations, it is crucial to compute the average transition moments over all orientations. For one-photon photodetachment using linearly polarized light, the orientationally averaged differential cross section is given by the following expression:³⁴

$$\frac{\partial\sigma}{\partial\Omega_{\mathbf{k}}} = \frac{\sigma}{4\pi} \left[1 + \beta \frac{3\cos^2\theta - 1}{2} \right], \quad (7)$$

where θ represents the angle between the direction of the photoelectron emission $\mathbf{k}$ and the polarization direction of the radiation $\bar{\epsilon}$. The total cross section can be derived from the differential cross sections at specific angles, specifically²⁵

$$\sigma = \frac{4\pi}{3} (\sigma_{par} + 2\sigma_{perp}), \quad (8)$$

where σ_{par} and σ_{perp} represent the differential cross sections when the angles are 0° and 90° , respectively. In this work, a high-precision full-space numerical integration method is introduced for orientation averaging.

σ_{par} and σ_{perp} are averaged over molecular orientations. [Figure 1](#) shows the experimental apparatus coordinate frame $C_{exp} = \{X, Y, Z\}$ and the molecular coordinate frame $C_{mol} = \{x, y, z\}$, where Z is aligned with $\bar{\epsilon}$.

The transformation from the experimental apparatus coordinates to the molecular coordinates can be accomplished using the direction cosine matrix T , in which c and s , respectively, represent the cosine and sine trigonometric functions, which is practically effective in orientational averaging of quantitative molecular parameters in the present work and has been applied as a routine in other fields, such as fluorescence, Raman, and pump-probe spectroscopies,³⁵

$$T = \begin{bmatrix} c\varphi c\theta c\chi - s\varphi s\chi & -c\varphi c\theta s\chi - s\varphi c\chi & c\varphi s\theta \\ s\varphi c\theta c\chi + c\varphi s\chi & -s\varphi c\theta s\chi + c\varphi c\chi & s\varphi s\theta \\ -s\theta c\chi & s\theta s\chi & c\theta \end{bmatrix}. \quad (9)$$

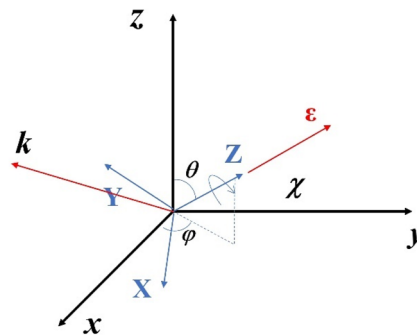


FIG. 1. Schematic of the experimental apparatus and molecular coordinate frame. The orientation of the molecule in the laboratory is described by Euler angles θ , φ , and χ . Here, θ represents the angle between the z-axis and the Z-axis, φ is the angle between the projection of the X-axis and the x-axis on the xy-plane, and χ denotes rotation around the Z-axis.

The square of the transition moment integral is integrated over the three Euler angles to perform an average over orientations,

$$\overline{|I|^2} = \frac{1}{8\pi^2} \int_0^\pi \int_0^{2\pi} \int_0^{2\pi} |I|^2 \sin\theta d\theta d\varphi d\chi. \quad (10)$$

For the triple integral of the above formula, this work first performs a spherical integration over θ and φ , followed by a one-dimensional integration over χ . For the one-dimensional integration, Simpson's rule is employed. For spherical integration, this study adopts the Lebedev numerical integration method.^{36–40}

Notably, for systems exhibiting limited anisotropy in their electronic wavefunctions, such as non-valence-bound anions or conventional anions that detach via non-valence channels,^{41,42} the orientation-averaging procedure can be recast in a coordinate-free form. By invoking the mild premise that the Dyson orbital's angular density is quasi-isotropic, the laboratory and molecular frames become congruent along three orthogonal axes, eliminating the need for explicit rotational averaging. Owing to the documented abundance of non-valence-bound anions and related detachment pathways,^{41–44} this frame-coincident approximation (FCA) offers a computationally streamlined yet justified alternative for such chemical scenarios.

We assume that the laboratory coordinate system coincides with the molecular coordinate system and then take the numerical average of the results in these three directions. [Figure 2](#) shows a schematic diagram with $\bar{\epsilon}$ aligned along the z-axis as an example. In this case, the total intensity can be derived exactly from the differential cross section at specific angles, where the expression for the transition dipole moment is given by Eq. (11). The final cross section is obtained by calculating the arithmetic mean of the cross sections in three orthogonal axes,

$$|I|_{par}^2 = \frac{k}{(2\pi)^3} \left| \left\langle e^{ikz} \right| \epsilon_z z \right| \psi_{dyson} \rangle \right|^2, \quad (11a)$$

$$|I|_{perp}^2 = \frac{1}{2} \frac{k}{(2\pi)^3} \left(\left| \left\langle e^{ikx} \right| \epsilon_z z \right| \psi_{dyson} \rangle \right|^2 + \left| \left\langle e^{iky} \right| \epsilon_z z \right| \psi_{dyson} \rangle \right|^2 \right). \quad (11b)$$

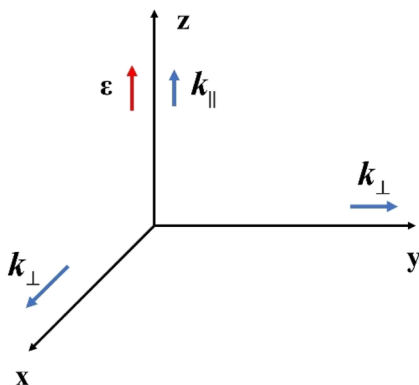


FIG. 2. Schematic diagram of the frame-coincident approximation (FCA) coordinate when $\vec{\epsilon}$ is consistent with the z axis; $k_{||}$ and $k_{\perp}$ indicate that k is parallel and perpendicular to $\vec{\epsilon}$, respectively.

III. COMPUTATIONAL DETAILS

The precision of the final result is closely relevant to the accuracy of the wavefunctions involved. In addition to the continuous wavefunction of the free electron, the anionic wavefunction of the target state also plays a crucial role in determining the PDCS. In DFT calculations, the choice of functional is crucial for accurately describing the wavefunctions of anions.^{45–48} Previous tests show that the CAM-B3LYP range-separated hybrid functional method⁴⁹ can accurately calculate photodetachment-related parameters in anionic systems,^{50,51} such as vertical detachment energy (VDE) and orbital energy. Therefore, geometric structures of the studied anions were computed using the CAM-B3LYP density function in this work.

In calculating the PDCS for anions, the inclusion of diffuse functions directly enhances the description of electron detachment; otherwise, the Gaussian basis function cannot exactly simulate the delocalized electron movement,^{52,53} leading to inaccuracies of the profile and magnitude of PDCS.²³ Moran *et al.* uncovered the existence of the basis set incompleteness error (BSIE) when adding diffuse functions to the basis without higher angular momentum functions, whereas the aug-cc-pVTZ basis set⁵⁴ provides the necessary BSIE balance.⁵⁵ Therefore, this work improves the aug-cc-pVTZ basis set by adding additional s and p diffuse functions until the PDCS values converge. The augmented basis set is denoted as aug-cc-pVTZ($n+$), where n indicates the additional layers of diffuse functions. The principle of adding diffuse functions follows the even-tempered method.⁵⁶ The number of additional diffuse layers (n) was determined separately for each anion via systematic convergence tests, since the spatial extent and diffuseness of the excess electron are system-dependent and thus vary among species. In practical calculations, for each system, n was increased until the PDCS changed by less than 1% when integrated over the photon-energy range of interest, ensuring a uniform convergence criterion across all anions.

The Dyson orbitals for anionic photodetachment processes were calculated at the EOM-CCSD/aug-cc-pVTZ($n+$)⁵⁷ level using the Q-Chem 6.0 program.⁵⁸ The DFT calculations were performed using the Gaussian 16 program,⁵⁹ while the PDCS calculations were conducted using a custom Python code.

TABLE I. Anion detachment energies and degeneracy factor used in this work. “*” represents the theoretical vertical detachment energy with the method of EOM-CCSD/aug-cc-pVDZ(1+).

Anion	Electron affinity (eV)	Degeneracy factor
H [−]	0.754 ^{60,61}	2
Li [−]	0.618 ^{62,63}	2
C [−]	1.260 ⁶⁴	3
O [−]	1.460 ⁶⁵	2
F [−]	3.400 ^{66,67}	6
Cl [−]	2.018 ⁶⁸	6
S [−]	2.010 ⁶⁹	2
OH [−]	1.830 ⁷⁰	4
NH ₂ [−]	0.744 ⁷¹	2
CN [−]	3.864 ⁷²	2
C ₂ H [−]	2.969 ^{73,74}	2
C ₇ H ₇ [−]	0.863 ⁷⁵	2
C ₂ H ₅ O ₂ N [−]	0.4095*	1
(H ₂ O) ₄ [−]	0.3030*	1

In treating photoionization, the total cross section is obtained by multiplying the single-electron cross section by the number of equivalent electrons and subsequently summing over all symmetry-allowed target states. Thus, the degeneracy factors should account for the degeneracy of molecular orbitals and the number of electrons occupying each orbital.^{11,26} For anionic atoms, the factor combines spin degeneracy (e.g., 2 for a closed shell) and the orbital angular-momentum degeneracy (e.g., 3 for a p -orbital). For anionic molecules, the factor reflects the occupancy of the detachment orbital and the spin multiplicity of the final neutral state, as determined by the electronic configuration.¹¹ The corresponding detachment energies and degeneracy factors of the tested anionic species (specifically for HOMO detachment) are listed in Table I.

IV. RESULTS AND DISCUSSION

A. Atomic anions

Figure 3 shows the calculated PDCS for the Li[−] in this study. Figure 3(a) presents the results at the CAM-B3LYP/aug-cc-pVTZ($n+$, $n = 0, 2$) level, with the dashed and dotted-dashed lines representing the length and momentum gauge results, respectively. Near the detachment threshold, the PDCS in the length gauge is approximately twice that in the momentum gauge. In theory, if the exact wavefunctions were available, the length and momentum gauge should yield consistent results, as they are simply different mathematical descriptions of the same photodetachment process. However, the introduction of approximations, such as the Hartree–Fock method, can compromise the gauge invariance (GI), often leading to significant discrepancies between the results obtained from different gauges.^{76,77} Although many studies have analyzed the primary differences between the two gauges,^{78–80} no definitive solution has yet been established. Instead, previous studies have assessed the accuracy of the wavefunction by comparing the differences in cross sections between the length and momentum gauge and used the combined results from both representations as the final result.¹⁵ In particular, the mixed-gauge PDCS is defined as

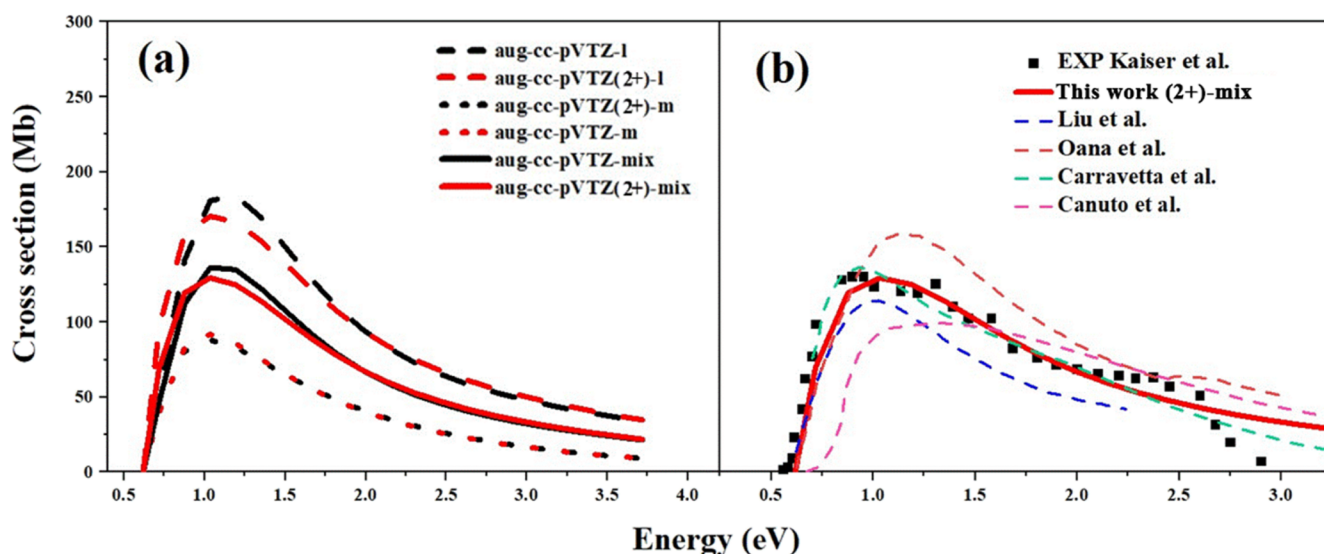


FIG. 3. Calculations of the PDCS for the Li^- using the CAM-B3LYP/aug-cc-pVTZ (black line) and CAM-B3LYP/aug-cc-pVTZ(2+) (red line) methods. (a) Results in the length (dashed line), momentum (dotted-dashed line), and mix (solid line) representations. (b) Comparison of the results of this work with experimental values and previous theoretical calculations, whose methods are as follows: B3LYP/AV5Z (Liu and Ning, blue)²⁵ reproduced from Liu and Ning, J. Chem. Phys. **143**, 144310 (2015), with the permission of AIP Publishing LLC; EOM-CCSD/6-311G(4+)(3pd) (Oana and Krylov, red)²³ reproduced from Oana and Krylov, J. Chem. Phys. **131**, 124114 (2009), with the permission of AIP Publishing LLC; MCLR/Stieltjes (Carravetta *et al.*, green)⁸¹ reproduced with permission from Carravetta *et al.*, J. Phys. B: At., Mol. Opt. Phys. **22**, 2133 (1989), copyright 1989 IOP Publishing; CCSDT1aPPA/Stieltjes (Canuto *et al.*, pink)⁸² reproduced with permission from Canuto *et al.*, J. Phys. B: At., Mol. Opt. Phys. **21**, 3891 (1988), copyright 1988 IOP Publishing; and experimental measurements from Kaiser *et al.*⁸³ reproduced with permission from Kaiser *et al.*, Z. Angew. Phys. **270**, 259 (1974), copyright 1974 Springer Nature.

the arithmetic mean of the length- and velocity-gauge cross sections: $\sigma_{\text{mix}} = \frac{1}{2}(\sigma^L + \sigma^V)$. Until further research identifies the optimal conditions for selecting one gauge over the other, the mixed results from both gauges provide a reasonable alternative and yield results closer to the experimental data. The solid line in the diagram represents this averaged result, labeled as “mix.”

Our tests indicated that after adding two layers of additional diffuse functions, the PDCS of the Li^- becomes convergent (see Fig. S1). As shown in Fig. 3(a), compared to the use of the aug-cc-pVTZ basis set alone, the main differences are observed in the peak position and magnitude of the cross section in the length gauge, while the momentum gauge results remain largely unchanged. This difference arises because the position operator $\hat{r}$ weights the long-range tail of the wavefunction more heavily, making it more sensitive to the description of diffuse electron density, whereas the momentum operator $\hat{p}$ depends more strongly on the local gradient near the nucleus. Therefore, the inclusion of additional diffuse functions is essential and has a substantial impact on the final PDCS results, particularly in the length gauge.

Figure 3(b) shows the comparison of the converged PDCS with the experimental measurements and previous theoretical calculations for Li^- .⁸³ The numerical average results from the length and momentum gauges in our study closely match the experimental data in terms of peak position and magnitude of the cross section, likely due to error cancellation between the two gauges. For the sake of simplicity and readability, the mixed gauge results will only be shown unless otherwise specified. The experimental values show a

sharp increase and decline at 2.5 eV, attributed to the influence of $\text{Li}(^2P)$ excited states,⁸³ which is successfully simulated by Ramsbottom *et al.*⁸⁴ and Gorczyca and Manson⁸⁵ with the R-matrix method. Previous theoretical studies have successfully replicated the profile of the PDCS for Li^- . Notably, Carravetta *et al.* used Slater-Type Orbital (STO) basis sets instead of Gaussian-Type Orbital (GTO) basis sets,⁸¹ and their calculations employing Multiconfigurational Linear Response (MCLR) and Stieltjes techniques closely matched experimental measurements.

Figures 4(a)–4(f) display the calculated PDCS for H^- , F^- , Cl^- , C^- , O^- , and S^- atomic anions in this study. H^- , which is the smallest anionic system, exhibits a PDCS profile similar to that of the Li^- anion, where photoelectrons are ejected from an s-orbital, characterized by a rapid rise followed by a decline. Compared with the experimental measurements by Smith and Burch,⁹⁰ the computational results from this study are generally consistent overall but underestimate the peak magnitudes and exhibit a more gradual decline.

For F^- , Cl^- , C^- , O^- , and S^- , where photoelectrons are ejected from a p-orbital, the threshold PDCSs for these anions exhibit a characteristic rapid rise followed by stabilization. For F^- , apart from the calculations by Oana and Krylov,²³ previous theoretical results are generally higher, whereas the results in this study are about 1.4 times that of the measured values by Mandl⁶⁸ and twice that of Vacqu   *et al.*⁹¹ The theoretical calculation results for Cl^- also show a significant overestimation. Unlike Cl^- and F^- , C^- and O^- are open-shell anions. The calculations by Liu *et al.* and Oana and Krylov²³

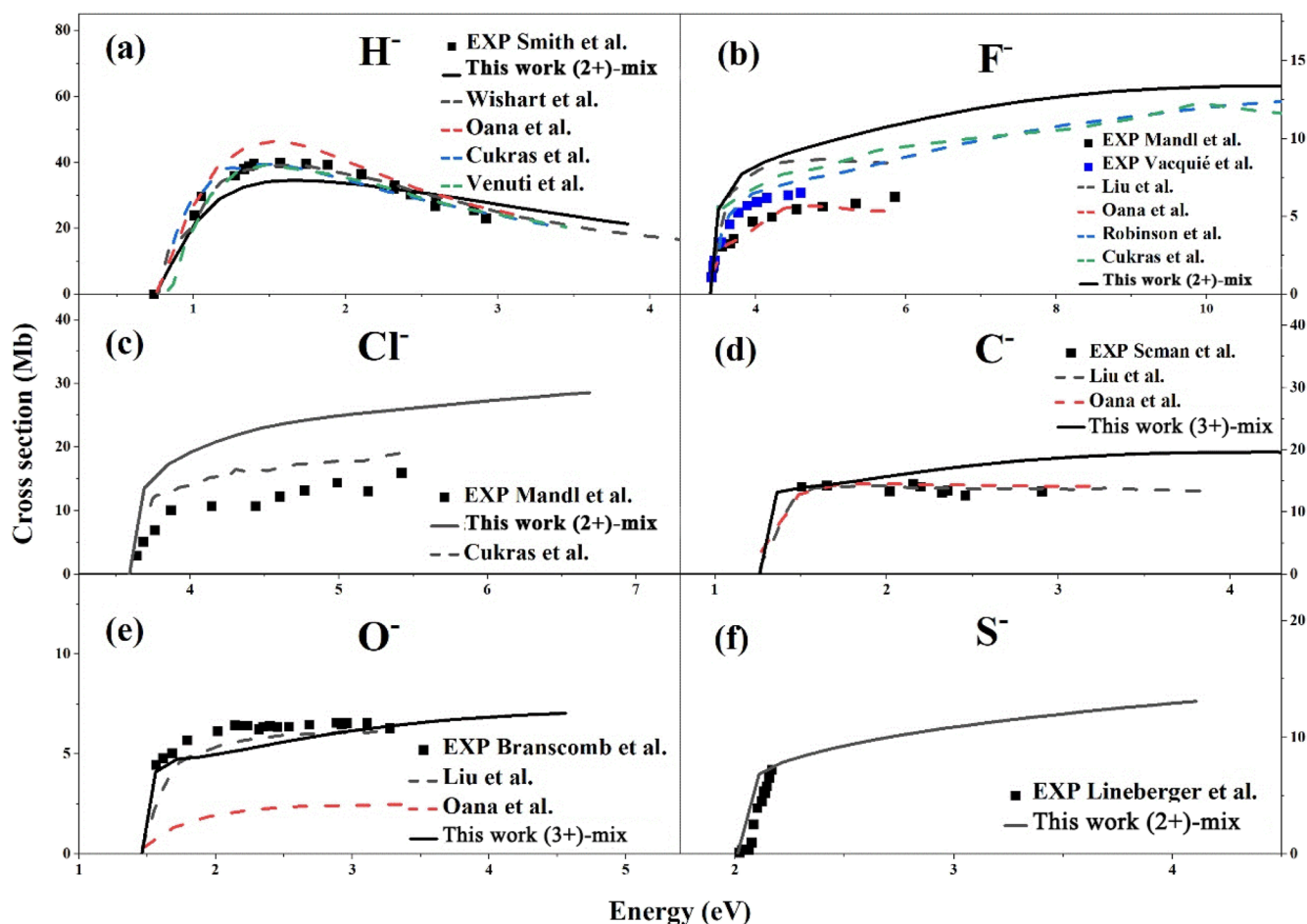


FIG. 4. Calculated PDCS for atomic anions at the CAM-B3LYP level compared with experiments and prior theories. In this work, (a) H^- , (b) F^- , (c) Cl^- , and (f) S^- used the aug-cc-pVTZ(2+) basis set, while (d) C^- and (e) O^- used the aug-cc-pVTZ(3+) basis set. Theoretical comparisons include the following: Wishart (the close-coupling method),⁸⁶ reproduced with permission from Wishart, *Mon. Not. R. Astron. Soc.* **187**, 59P (1979), licensed under a Creative Commons Attribution (CC BY) license; Oana and Krylov [EOM-CCSD/6-311G(4+)(3pd)],²³ reproduced from Oana and Krylov, *J. Chem. Phys.* **131**, 124114 (2009), with the permission of AIP Publishing LLC; Cukras *et al.* (Lanczos CCSD pseudo-spectra/Stieltjes),⁸⁷ reproduced from Cukras *et al.*, *J. Chem. Phys.* **141**, 174315 (2014), with permission from AIP Publishing LLC; Venuti and Decleva (configuration interaction/B-spline),⁸⁸ reproduced with permission from Venuti and Decleva, *J. Phys. B: At., Mol. Opt. Phys.* **30**, 4839 (1997), copyright 1997 IOP Publishing; Liu and Ning (B3LYP/AV5Z),²⁵ reproduced from Liu and Ning, *J. Chem. Phys.* **143**, 144310 (2015), with the permission of AIP Publishing LLC; and Robinson and Geltman (the central-field model),⁸⁹ reproduced with permission from Robinson and Geltman, *Phys. Rev.* **153**, 4 (1967), copyright 1967 American Physical Society. Experimental measurements from (a) Smith and Burch,⁹⁰ reproduced with permission from Smith and Burch, *Phys. Rev.* **116**, 1125 (1959), copyright 1959 American Physical Society; (b) Mandl,⁶⁸ reproduced with permission from Mandl, *Phys. Rev. A* **14**, 345 (1976), copyright 1976 American Physical Society, and Vacquie *et al.*,⁹¹ reproduced with permission from Vacquie *et al.*, *Phys. Rev. A* **35**, 1615 (1987), copyright 1987 American Physical Society; (c) Mandl,⁶⁸ reproduced with permission from Mandl, *Phys. Rev. A* **14**, 345 (1976), copyright 1976 American Physical Society; (d) Seman and Branscomb,⁹² reproduced with permission from Seman and Branscomb, *Phys. Rev.* **125**, 1602 (1962), copyright 1962 American Physical Society; (e) Branscomb *et al.*,⁹³ reproduced from Branscomb *et al.*, *J. Chem. Phys.* **43**, 2906 (1965), with permission from AIP Publishing LLC; and (f) Lineberger and Woodward,⁶⁹ reproduced with permission from Lineberger and Woodward, *Phys. Rev. Lett.* **25**, 424 (1970), copyright 1970 American Physical Society.

align well with the experimental results of the C^- PDCS measured by Seman and Branscomb.⁹² For O^- , the calculations by this study and Liu and Ning²⁵ are consistent with the experimental values obtained by Branscomb *et al.*,⁹³ with this study capturing the rapid rise near the threshold well. In contrast, calculations by Oana and Krylov²³ show a significant deviation from experimental values. In addition, the theoretical PDCS for S^- is presented for the first time in this

work. The calculated results show good agreement with experimental values. The theoretical calculations indicate that the PDCS of the S anion will level off around 2.2 eV.

Overall, the PDCSs of s-orbital anions (Li^- and H^-) and p-orbital anions (F^- , Cl^- , C^- , O^- , and S^-) were computed using the newly proposed approach in this study. The mixed gauge results show good agreement with the previous experimental and

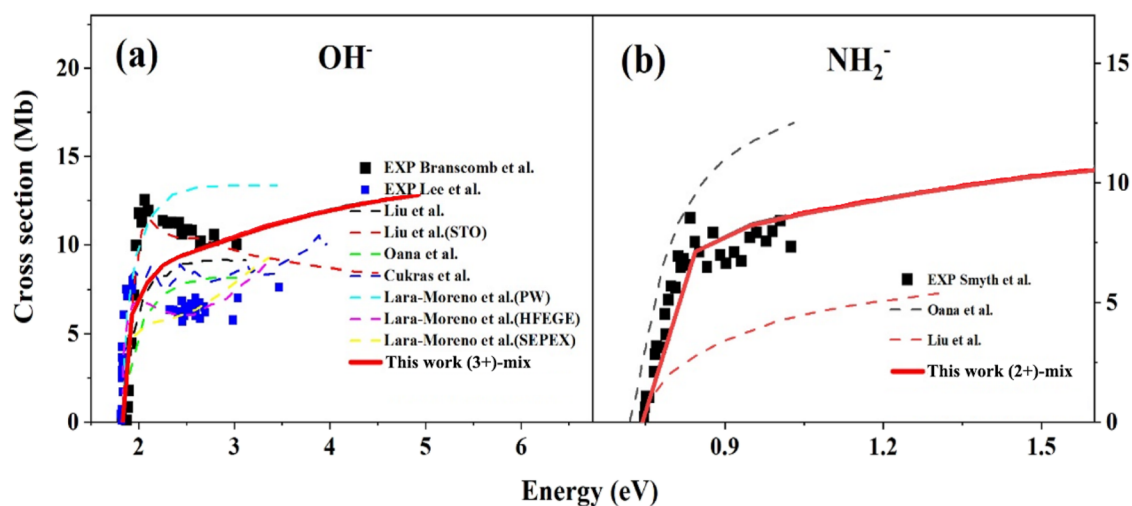


FIG. 5. Calculated PDCS for molecular anions at the CAM-B3LYP level compared with experiments and prior theories. This work includes aug-cc-pVTZ(3+) for (a) OH^- and aug-cc-pVTZ(2+) for (b) NH_2^- . Theoretical comparisons include the following: (a) For OH^- , Liu *et al.* (B3LYP/AV6Z and B3LYP/QZ3P-2d methods),²⁵ reproduced from Liu and Ning, J. Chem. Phys. **143**, 144310 (2015), with the permission of AIP Publishing LLC; Oana and Krylov [EOM-CCSD/6-311G(4+,4+)(3df,3pd)],²³ reproduced from Oana and Krylov, J. Chem. Phys. **131**, 124114 (2009), with the permission of AIP Publishing LLC; Cukras *et al.* (Lanczos CCSD pseudo-spectra/Stieltjes),⁸⁷ reproduced from Cukras *et al.*, J. Chem. Phys. **141**, 174315 (2014), with the permission of AIP Publishing LLC; and Lara-Moreno *et al.* (single-center approach, where the continuum state functions were described using PW, HFEGE, and SEPEX, respectively),¹¹ reproduced with permission from Lara-Moreno *et al.*, Phys. Rev. A **99**, 033412 (2019), copyright 2019 American Physical Society. (b) For NH_2^- , Oana and Krylov [EOM-CCSD/6-311G(3+,3+)(3df,3pd)],²³ reproduced from Oana and Krylov, J. Chem. Phys. **131**, 124114 (2009), with the permission of AIP Publishing LLC, and Liu and Ning (B3LYP/AV6Z),²⁵ reproduced from Liu and Ning, J. Chem. Phys. **143**, 144310 (2015), with the permission of AIP Publishing LLC. Experimental measurements from (a) Branscomb,⁷⁰ reproduced with permission from Branscomb, Phys. Rev. **148**, 11 (1966), copyright 1966 American Physical Society and reproduced from Lee and Smith, J. Chem. Phys. **70**, 1727 (1979), with the permission of AIP Publishing LLC, and (b) Smyth and Brauman,⁷¹ reproduced from Smyth and Brauman, J. Chem. Phys. **56**, 4620 (1972), with the permission of AIP Publishing LLC.

theoretical data. The next step will be to validate these methodologies through computational research on the PDCS of a series of molecular anions.

B. Molecular anions

Figure 5(a) displays the theoretical calculations of the PDCS for the OH^- . Two sets of experimental data are available, with measurements by Branscomb⁷⁰ being higher than those of Lee and Smith,⁹⁴ partly attributable to differences in instrumental conditions. Several research groups have used various methods to calculate the PDCS for OH^- . Liu and Ning,²⁵ using both GTO and STO basis functions, found that STO basis functions better reproduce the peak at 2 eV observed in Branscomb *et al.*'s experiments, while GTO calculations showed a rapid rise followed by stabilization. Notably, the differences between the GTO and STO bases in other systems are minimal in Liu *et al.*'s study, likely due to the unique wavefunction of the OH^- . Lara-Moreno *et al.*¹¹ employed the plane wave (PW), Hara's free-electron-gas exchange potential (HFEGE),⁹⁵ and separable exchange potential (SEPEX)⁹⁶ to describe the continuum wavefunctions of the outgoing electrons, with the latter two potentials incorporating exchange-correlation effects. Their results indicate that including these effects improves agreement with Lee *et al.*'s experimental data. Lara-Moreno *et al.*'s PW-based calculations showed trends similar to those of this study, but with higher magnitudes, possibly due to their failure to average gas-phase molecular orientations. The differences between PW, HFEGE,

and SEPEX methods highlight the significant impact of neglecting photoelectron-molecular interactions on the PDCS profile for polar systems such as the OH^- . Our approach agrees with other results at the threshold, whereas the trend diverges at higher energy due to the use of the plane wave approximation and GTOs.

Figure 5(b) shows the calculated PDCS for NH_2^- . Since Smyth and Brauman measured relative PDCS,⁷¹ the experimental values were linearly scaled to match the theoretical results in this study for comparison. The trends in the calculations from this study and previous theoretical predictions are consistent with experimental values. However, in Liu and Ning's study,²⁵ the degeneracy factor was set to 1 instead of 2, which accounts for the difference in magnitude.

Figure 6(a) displays our theoretical calculations for CN^- . Experimental measurements on the PDCS of CN^- are very limited, with the only available measurement by Kumar *et al.* at 4.66 eV.¹ Our results align well with Kumar *et al.*'s data. Compared to Brigg's study,²⁰ our results are slightly overestimated, but the trends are similar. As shown in Fig. 6(b), within the energy range of interest, the photodetachment of CN^- involves electrons from the HOMO, HOMO-1, and HOMO-2, corresponding to three different final states of the CN radical. The main differences between our calculations and those of Brigg arise from the discrepancies in the cross sections of the $A^2\Pi$ and $B^2\Sigma^-$ photodetachment channels.

Figure 7(a) presents the PDCS calculations for C_2H^- . Previous studies indicate that the ground state electronic configuration of the C_2H^- is $4\sigma^2 1\pi^4 5\sigma^2$,^{72,73,97} which can be correctly predicted

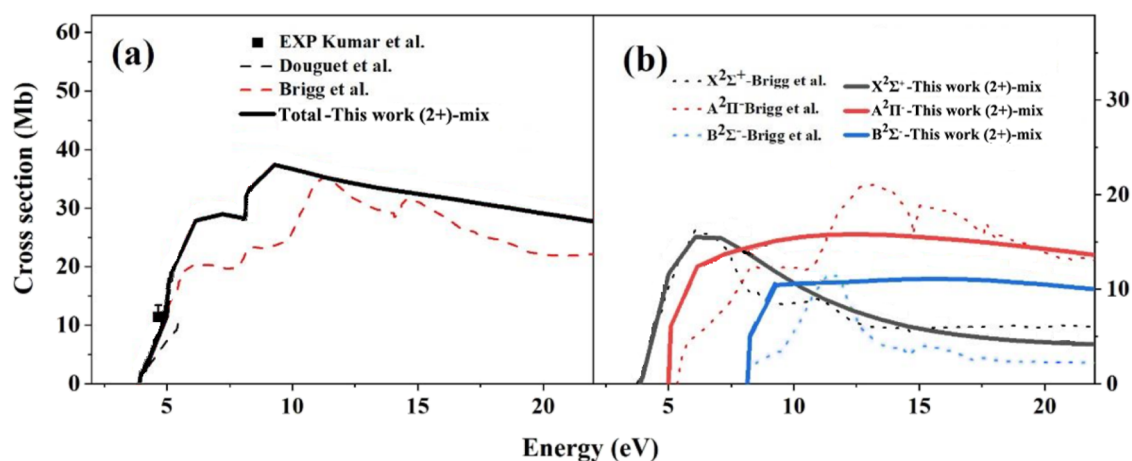


FIG. 6. (a) Total and (b) single-channel PDCS for the CN^- calculated using CAM-B3LYP/aug-cc-pVTZ(2+), with comparisons with the complex Kohn variational method used by Douguet *et al.*, reproduced with permission from Douguet *et al.*, Phys. Rev. A **88**, 052710 (2013), copyright 2013 American Physical Society, and the R-matrix method used by Brigg, reproduced with permission from Brigg, "R-matrix calculations of electron-molecule collisions: Flexible implementation and practice" (University College London, 2018), licensed under a Creative Commons Attribution (CC BY) license. Experimental measurements from Kumar *et al.*,¹ reproduced with permission from Kumar *et al.*, Astrophys. J. **776**, 25 (2013), licensed under a Creative Commons Attribution (CC BY) license.

using the B3LYP⁹⁸ functional. The experimental PDCS values measured by Best *et al.* are concentrated near the threshold,⁷² with our theoretical results being slightly overestimating these values. Similar to the CN^- results, our calculations align closely with Brigg's results,²⁰ but are slightly higher than those by Douguet *et al.*³ As shown in Fig. 7(b), within the energy range of interest, photodetachment of C_2H^- involves electrons from the HOMO and HOMO-1, corresponding to the $X^2\Sigma^+$ and $A^2\Pi$ final states of the C_2H radical. In single-channel PDCS calculations, our results qualitatively

align with those of Brigg,²⁰ despite minor discrepancies (resonance structures) arising from their use of the more sophisticated R-matrix method—an electron-scattering formalism that transcends the plane wave approximation.

Following the standards established in the Leiden PICS/PDCS database,^{99,100} the accuracy of interstellar atomic and molecular data is classified into four levels: A^+ (<20%) and A (<30%) levels represent deviations smaller than intrinsic experimental uncertainties, rendering them negligible. In contrast, B and C levels denote

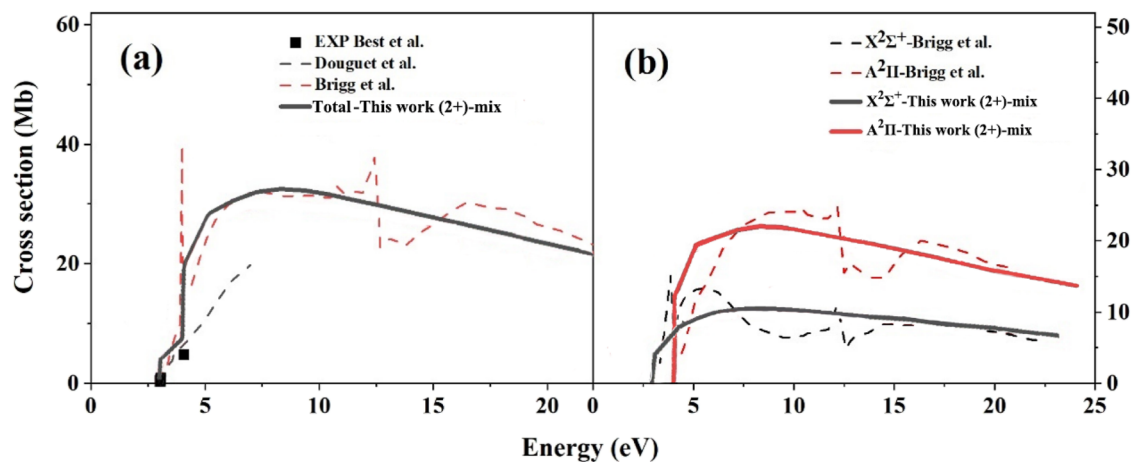


FIG. 7. (a) Total PDCS for the C_2H^- calculated using B3LYP/aug-cc-pVTZ(2+), with comparative analyses against the complex Kohn variational method employed by Douguet *et al.*, reproduced with permission from Douguet *et al.*, Phys. Rev. A **90**, 063410 (2014), copyright 2014 American Physical Society, and the R-matrix method used by Brigg, reproduced with permission from Brigg, "R-matrix calculations of electron-molecule collisions: Flexible implementation and practice" (University College London, 2018), licensed under a Creative Commons Attribution (CC BY) license. (b) Single-channel PDCS for the C_2H^- . Experimental measurements from Best *et al.*,⁷² reproduced with permission from Best *et al.*, Astrophys. J. **742**, 63 (2011), licensed under a Creative Commons Attribution (CC BY) license.

accuracies within factors of 2 and 10, respectively. According to the above criteria, the present algorithm predicts the PDCS of Li^- at the A^+ -level, which is at the same level as that predicted by Carravetta *et al.* and superior to others (see Fig. 3 for details). For most anions shown in Figs. 4–7, our method again belongs to the A^+ -level. For anions F^- and Cl^- , our method becomes the B-level because the over-simplified single-particle approximation neglects the electron correlation and exchange effects. Nevertheless, it should be emphasized that our algorithm consistently delivers reliable accuracy across both atomic and molecular anions, demonstrating its robustness and thus highlighting the enhancement in accuracy.

C. Computational efficiency and accuracy on larger molecule anions

Recent studies have increasingly focused on large molecular anions and their unique roles in photo-dominated regions (PDRs),^{6,100–102} including polycyclic aromatic hydrocarbons (PAHs)¹⁰³ and linear carbon-chain anions.¹⁰⁴ However, as the number and variety of atoms increase, the wavefunctions of these systems become more complex. Consequently, the contribution of higher angular momentum components in the partial plane wave expansion cannot be neglected. In this section, we evaluate the effectiveness of our approach by testing several representative systems.

1. Benchmarking and computational efficiency

Figure 8(a) presents the mixed gauge results compared with the linear scaled relative cross section experimental data for C_7H_7^- .⁷⁵ Apart from the anomalous points before the detachment threshold (0.863 eV), the trend of our approach is consistent with

experimental values. Figure 8(b) presents single-channel length gauge results at higher energies alongside traditional plane wave expansion results from the ezDyson software.^{24,26} Literally, the only difference between these two approaches lies in the use of the plane wave expansion. In practice, increasing photoelectron energy requires higher angular momentum terms for convergence in the expansion method. Within the tested scope, the truncation at an angular quantum number of at least 6 is necessary, while our results consistently coincide with the convergent expansion results. Considering multiple detachment channels, obtaining reliable results from the expansion method becomes more challenging, and further tests on the truncation criteria are warranted.

Moreover, the CPU time consumption of each method was summarized to assess computational efficiency for C_7H_7^- (shown in Table II). In particular, the reported CPU-time comparisons refer exclusively to the integral-evaluation step, not to the preceding quantum chemical orbital calculation. Therefore, the speed-up factor reflects the efficiency gain of our recursive Gaussian-integral scheme over conventional plane wave expansion for a given set of orbitals, regardless of how those orbitals were obtained by the electronic structure method. Eleven calculation points were sampled at 1 eV intervals over an electron kinetic energy range of 0.1–10.1 eV. For the orientation average at each point, the full-space method was adapted, where Lebedev points were 50. The computational cost of the conventional angular momentum truncation method scales quadratically with l_{max} (Fig. S2), leading to prohibitively long execution times for complex systems, where high l_{max} is required, while the present analytical algorithm does not. For the recommended l_{max} of 5, our analytical method is around 8 times faster. Strikingly, the analytical algorithm is faster than the truncated method even for the smallest l_{max} , which is far away from the convergence. This

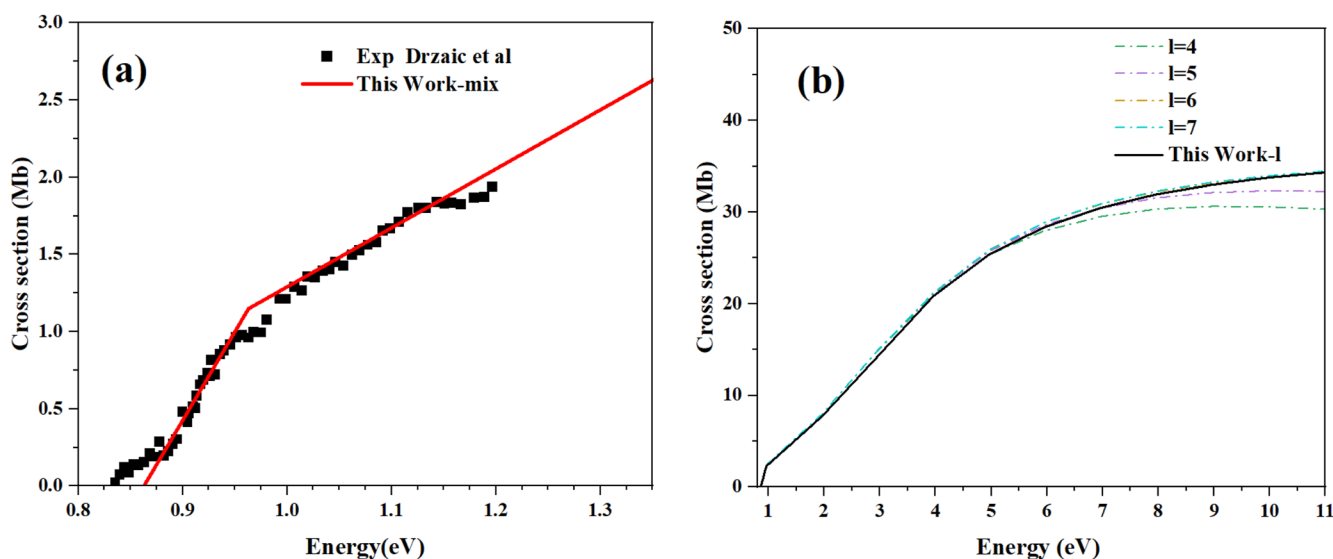


FIG. 8. PDCS for the benzyl anion calculated using EOM-CCSD/aug-cc-pVTZ(2+). (a) Threshold mix gauge results with previous experiment relative measurements by Drzaic and Brauman⁷⁵ (scaled by a factor of 2),⁷⁵ reproduced with permission from Drzaic and Brauman, *J. Phys. Chem.* **88**, 5285 (1984), copyright 1984 American Chemical Society. (b) Single channel length gauge results with comparisons with the angular momentum truncation method from ezDyson (l represents maximum angular momentum).

TABLE II. CPU time consumption (unit: s) for $C_7H_7^-$. The first row corresponds to our method, while the remaining rows correspond to the angular momentum truncation method from ezDyson, where $l_{\max}$ represents the maximum angular momentum number. All calculations were performed at the method of EOM-CCSD/aug-cc-pVTZ(2+).

Computing method	CPU time (s)
This work	297
$l_{\max} = 1$	425
$l_{\max} = 2$	671
$l_{\max} = 3$	1067
$l_{\max} = 4$	1483
$l_{\max} = 5$	2360
$l_{\max} = 6$	3079

result should be attributed to the recursive solution algorithm, which directly evaluates transition moments in Cartesian Gaussian form without performing any angular-momentum decomposition. For systems with more complex wavefunctions and higher photoelectron energies, inclusion of higher angular momentum components further amplifies the efficiency gap between the two methods. In summary, our approach circumvents the limitations inherent in the plane wave expansion and demonstrates significant potential for accurately and efficiently calculating PDCS of large, complex molecular anions.

2. Validation and application with frame-coincident approximation

As discussed in Sec. II C, the frame-coincident approximation (FCA) is intended for molecular anions exhibiting low anisotropy in their wavefunctions. Validation integrated all prior systems (Table

TABLE III. CPU time consumption (unit: s) for two anions calculated using our two methods ("full-rotational" represents the full-space numerical integration method, while "FCA" represents calculation with the frame-coincident approximation. All calculations were performed at the method of EOM-CCSD/aug-cc-pVDZ(1+).

Species	Orientation method	
	Full-rotational	FCA
Glycine ⁻	520	51.8
(H ₂ O) ₄ ⁻	614	53.9

SI) with two dipole-bound prototypes: the glycine anion (glycine- A^-)⁴¹ and (H₂O)₄⁻ anion (4Rf⁻).⁴² representing biomolecular and cluster anions, respectively. Notably, research on water cluster anions has intensified, particularly in the context of photoelectron spectroscopy detection and dynamics.^{105–109} Figure 9 assesses the fidelity of each gauge representation when the FCA is employed vs full-space averaging for these anions. To our knowledge, no experimental or reliable theoretic adiabatic detachment energies have been reported for two systems; thus, vertical detachment energies from EOM-CCSD are used as reference points for PDCS. Across every gauge, the FCA approach exhibits excellent consistency with the full-space integration method, while reducing computational time by an order of magnitude (see Table III). Therefore, the FCA provides a robust and efficient approximation for systems such as dipole-bound anions.

Critically, gauge discrepancies—manifest in magnitude, peak positions, and spectral topology (Figs. 9 and S3)—persist pervasively across studied systems. For instance, Fig. S3 reveals the systematic deviations transcending stochastic errors among all tested organic molecular anions in the 0–10 eV photoelectron range. This energy

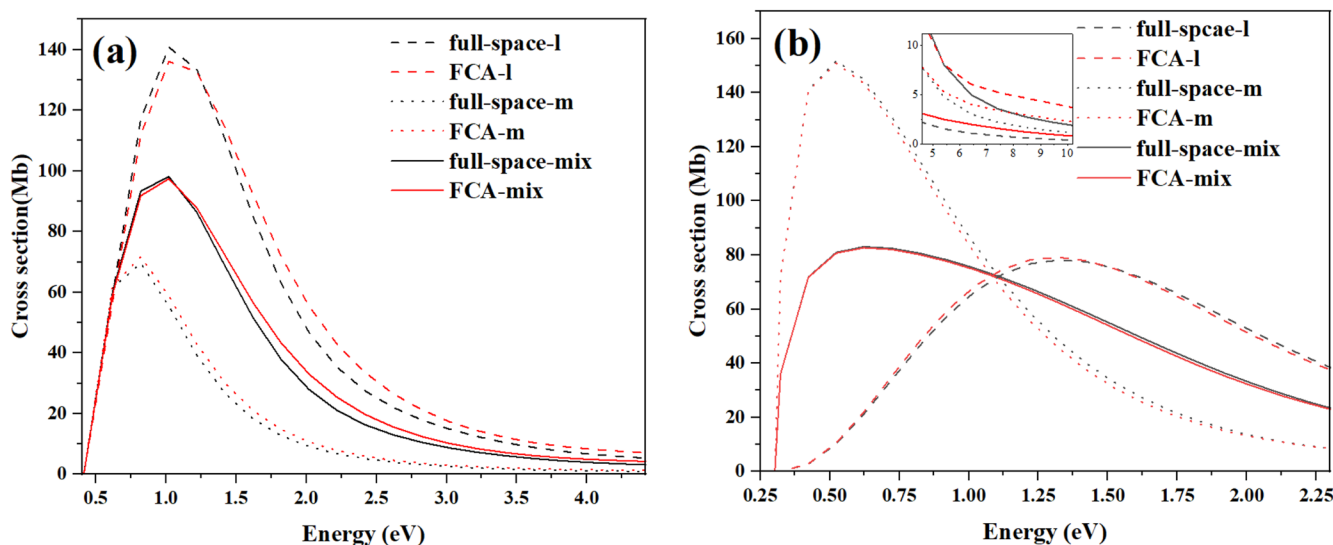


FIG. 9. PDCS for (a) glycine anion and (b) (H₂O)₄⁻ calculated using EOM-CCSD/aug-cc-pVDZ(1+). The black and red lines correspond to the full-space numerical integration and the FCA method. These methods are depicted in three different representations: length (dashed line), momentum (dotted line), and mixed (solid line).

regime governs the VUV-driven dynamics in photochemistry¹¹⁰ and the astrochemical processes in the PDR,¹⁰⁰ highlighting the urgent need to delineate gauge-validity domains. Prevailing single-gauge methodologies risk unreliability when their representations mismatch physical contexts. While mixed approaches combining length and momentum gauge results have been shown to mitigate errors, this strategy remains largely empirical rather than rigorously justified.

V. CONCLUSION

This study presents a novel method for calculating transition moment integrals analytically, utilizing recursive two-center Gaussian integral expressions based on Cartesian Gaussian basis functions. This approach combines plane waves with complex Gaussian integrals to overcome the convergence difficulties encountered in conventional plane wave expansion techniques. By accounting for the random orientation of gas-phase molecules, we derive the total PDCS for anionic systems. Our results indicate that the electron cloud distribution of anions is dispersed, underscoring the necessity of incorporating sufficient diffuse functions in the basis set to achieve convergent PDCS. Moreover, discrepancies in PDCS results between length and velocity gauges necessitate rigorous scrutiny of gauge selection in theoretical representations.

In this work, we employ a full-space numerical integration method, coupled with a direction transformation matrix, to average molecular orientation. This technique is particularly effective for polyatomic systems and demonstrates high efficiency in computing PDCS. For molecular anions with low anisotropy wavefunctions, the requisite full-space numerical integration can collapse onto three mutually orthogonal axes, delivering computational savings of roughly an order of magnitude without loss of accuracy.

The computational framework was tested on atomic anions, such as H^- , Li^- , C^- , O^- , F^- , S^- , and Cl^- , as well as molecular anions, such as OH^- , NH_2^- , CN^- , and C_2H^- . The results demonstrate good consistency with existing experimental and theoretical values, verifying the effectiveness and accuracy of our method. The method's high efficiency is further demonstrated on larger anionic molecules, such as C_7H_7^- , making it well-suited for large-scale, computationally intensive systems. In addition, the frame-coincident approximation was validated through test cases involving glycine $^-$ and $(\text{H}_2\text{O})_4^-$, significantly reducing computational costs for non-valence bound anions.

In summary, the computational framework introduced in this study provides a robust and efficient tool for calculating PDCS across a broad spectrum of anionic systems, paving the way for more accurate and computationally feasible investigations of complex molecular anions in the future.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for information about recursive formalism for two-center Gaussian integrals, basis set convergence analysis for Li^- , computational efficiency improvement relative to plane wave expansion at given maximum angular momentum, accuracy assessment of the FCA orientation averaging model, and gauge-dependence analysis on a series of molecular anions.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jiabin Huang and Dongyang Li contributed equally to this work.

Jiabin Huang: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal). **Dongyang Li:** Data curation (equal); Formal analysis (equal); Writing – original draft (equal). **Guangda Luo:** Data curation (equal); Validation (equal). **Mengqi Wu:** Data curation (equal); Validation (equal). **Sai Duan:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Feng Zhang:** Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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